

The United States Navy continued filling armour-piercing shells with explosive D after some other nations had switched to TNT, but began filling naval mines, bombs, depth charges, and torpedo warheads with burster charges of crude grade B TNT with the color of brown sugar and requiring an explosive booster charge of granular crystallized grade A TNT for detonation. High-explosive shells were filled with grade A TNT, which became preferred for other uses as industrial chemical capacity became available for removing xylene and similar hydrocarbons from the toluene feedstock and other nitrotoluene isomer byproducts from the nitrating reactions.^[10]



Chunks of explosives-grade TNT



Trinitrotoluene melting at 81 °C (178 °F)

ChEMBL	ChEMBL1236345 (https://www.ebi.ac.uk/chembl/explorable/compound/ChEMBL1236345) ✗
ChemSpider	8073 (https://www.chemspider.com/Chemical-Structure.8073.html) ✗
DrugBank	DB01676 (https://www.drugbank.ca/drugs/DB01676) ✓
ECHA InfoCard	100.003.900 (https://echa.europa.eu/substance-information/-/substanceinfo/100.003.900)
EC Number	204-289-6
KEGG	C16391 (https://www.kegg.jp/entry/C16391) ✗
PubChem CID	8376 (https://pubchem.ncbi.nlm.nih.gov/compound/8376)
RTECS number	XU0175000
UNII	H43RF5TRM5 (https://precision.fda.gov/uniisearch/srs/unii/H43RF5TRM5) ✓
UN number	0209 – Dry or wetted with < 30% water 0388, 0389 – Mixtures with trinitrobenzene, hexanitrostilbene
CompTox Dashboard (EPA)	DTXSID7024372 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID7024372)
InChI	InChI=1S/C7H5N3O6/c1-4-6(9(13)14)2-5(8(11)12)3-7(4)10(15)16/h2-3H,1H3 ✗ Key: SPSSULHKWOKEEL-UHFFFAOYSA-N ✗
InChI	InChI=1/C7H5N3O6/c1-4-6(9(13)14)2-5(8(11)12)3-7(4)10(15)16/h2-3H,1H3
InChI	InChI=1/C7H5N3O6/c1-4-2-3-5(8(11)12)7(10(15)16)6(4)9(13)14/h2-3H,1H3 Key: FPKOPBFLPLFWAD-UHFFFAOYAR
SMILES	<chem>Cc1c(cc(cc1[N+](=O)[O-])[N+](=O)[O-])[N+](=O)[O-]</chem>



M795 artillery shells with fuzes fitted, labelled to indicate a filling of TNT

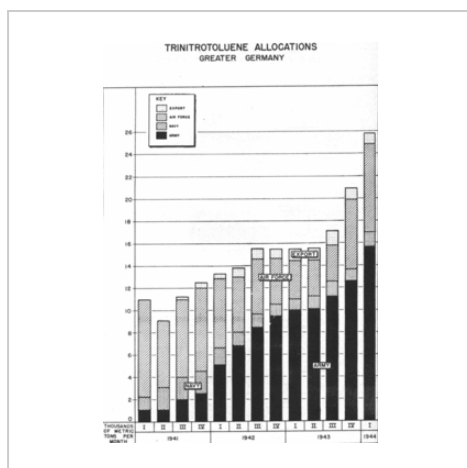


M107 artillery shells. All are labelled to indicate a filling of "Comp B" (mixture of TNT and RDX) and have fuzes fitted



A group of M120 Rak mortar shells. The dark green shells on the left are stencilled to indicate a filling of TNT

Properties	
Chemical formula	$C_7H_5N_3O_6$
Molar mass	227.132 g·mol ⁻¹
Appearance	Pale yellow solid crystalline needles, flakes or prills
Density	1.654 g/cm ³
Melting point	80.35 °C (176.63 °F; 353.50 K)
Boiling point	240.0 °C (464.0 °F; 513.1 K) (decomposes) ^[2]
Solubility in water	0.13 g/L (20 °C)
Solubility in ether, acetone, benzene, pyridine	soluble
Vapor pressure	0.0002 mmHg (20°C) ^[3]
Explosive data	
Shock sensitivity	Insensitive
Friction sensitivity	Insensitive to 353 N
Detonation velocity	6900 m/s
RE factor	1.00
Hazards	
GHS labelling:	
Pictograms	
Signal word	Danger
Hazard statements	H201, H301, H311, H331, H373, H411
Precautionary statements	P210, P273, P309+P311, P370+P380, P373, P501
NFPA 704 (fire diamond)	
Lethal dose or concentration (LD, LC):	
LD ₅₀ (median dose)	795 mg/kg (rat, oral) 660 mg/kg (mouse, oral) ^[4]
LD _{Lo} (lowest published)	500 mg/kg (rabbit, oral) 1850 mg/kg (cat, oral) ^[4]
NIOSH (US health exposure limits):	
PEL (Permissible)	TWA 1.5 mg/m ³ [skin] ^[3]

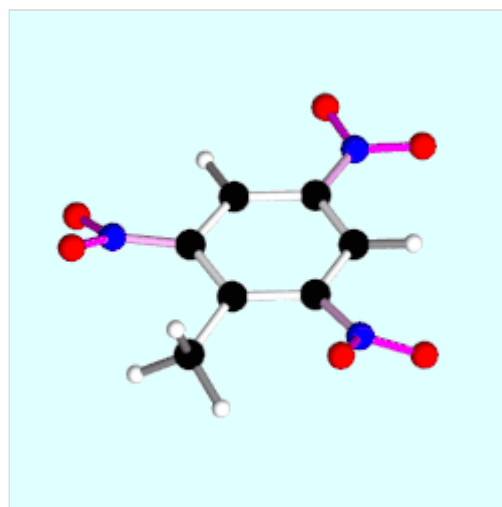


Analysis of TNT production by branch of the German armed forces between 1941 and the first quarter of 1944, shown in thousands of tons per month



Detonation of the 500-ton TNT explosive charge as part of Operation Sailor Hat in 1965. The passing blast-wave left a white water surface behind and a white condensation cloud is visible overhead.

REL (Recommended)	TWA 0.5 mg/m ³ [skin] ^[3]
IDLH (Immediate danger)	500 mg/m ³ ^[3]
Safety data sheet (SDS)	ICSC 0967 (http://www.inchem.org/documents/icsc/icsc/eics0967.htm)
Related compounds	
Related compounds	picric acid hexanitrobenzene 2,4-Dinitrotoluene
Except where otherwise noted, data are given for materials in their <u>standard state</u> (at 25 °C [77 °F], 100 kPa). ✗ <u>verify</u> (what is ✓✗ ?) <u>Infobox references</u>	



Trinitrotoluene spin view

Preparation

In industry, TNT is produced in a three-step process. First, toluene is nitrated with a mixture of sulfuric and nitric acid to produce mononitrotoluene (MNT). The MNT is separated and then renitrated to dinitrotoluene (DNT). In the final step, the DNT is nitrated to trinitrotoluene (TNT) using an anhydrous mixture of nitric acid and fuming sulfuric acid (oleum). Nitric acid is consumed by the manufacturing process, but the diluted sulfuric acid can be reconcentrated and reused.

After nitration, TNT can either be purified by crystallization from an organic solvent or stabilized by a process called sulfitation, where the crude TNT is treated with aqueous sodium sulfite solution to remove less stable isomers of TNT and other undesired reaction products. The rinse water from sulfitation is

known as red water and is a significant pollutant and waste product of TNT manufacture.^[11]

Control of nitrogen oxides in feed nitric acid is very important because free nitrogen dioxide can result in oxidation of the methyl group of toluene. This reaction is highly exothermic and carries with it the risk of a runaway reaction leading to an explosion.

In the laboratory, 2,4,6-trinitrotoluene is produced by a two-step process. A nitrating mixture of concentrated nitric and sulfuric acids is used to nitrate toluene to a mixture of mono- and di-nitrotoluene isomers, with careful cooling to maintain temperature. The nitrated toluenes are then separated, washed with dilute sodium bicarbonate to remove oxides of nitrogen, and then carefully nitrated with a mixture of fuming nitric acid and sulfuric acid.

X-ray crystallography determined that each of the three planar nitro groups is substantially rotated out of the plane of the benzene ring.^[12]

Applications

TNT is one of the most commonly used explosives for military, industrial, and mining applications. TNT has been used in oil and gas fracking (hydraulic fracturing) in shale formations. The technique involves displacing and detonating nitroglycerin in hydraulically induced fractures followed by wellbore shots using pelletized TNT.^[13]

TNT is valued partly because of its insensitivity to shock and friction, with reduced risk of accidental detonation compared to more sensitive explosives such as nitroglycerin. TNT melts at 80 °C (176 °F), far below the temperature at which it will spontaneously detonate, allowing it to be poured or safely combined with other explosives. TNT neither absorbs nor dissolves in water, which allows it to be used effectively in wet environments. To detonate, TNT must be triggered by a pressure wave from a starter explosive, called an explosive booster.^[14]

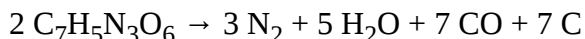
Although blocks of TNT are available in various sizes (e.g., 250 g, 500 g, 1,000 g), it is more commonly encountered in synergistic explosive blends comprising a variable percentage of TNT plus other ingredients. Examples of explosive blends containing TNT include:

- Amatex (ammonium nitrate and RDX)^[15]
- Amatol (ammonium nitrate^[16])
- Baratol (67% barium nitrate and 33% TNT)^[17]
- Composition B (RDX and paraffin wax^[18])
- Composition H6
- Cyclotol (RDX)^[19]
- Ednatol
- Hexanite (hexanitrodiphenylamine^{[20][21]})
- Minol (40% TNT, 40% ammonium nitrate, 20% aluminium^[17])
- Octol (75% HMX and 25% TNT^[17])
- Pentolite
- Picratol
- Tetrytol

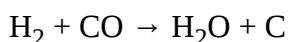
- Torpex
- Tritonal

Explosive character

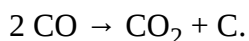
Upon detonation, TNT undergoes a decomposition equivalent to the reaction



plus some of the reactions



and



The reaction is exothermic but has a high activation energy in the gas phase (~62 kcal/mol). The condensed phases (solid or liquid) show markedly lower activation energies of roughly 35 kcal/mol due to unique bimolecular decomposition routes at elevated densities.^[22] Because of the production of carbon, TNT explosions have a sooty appearance. Because TNT has an excess of carbon, explosive mixtures with oxygen-rich compounds can yield more energy per kilogram than TNT alone. During the 20th century amatol, a mixture of TNT with ammonium nitrate, was a widely used military explosive.^[23]

TNT can be detonated with a high velocity initiator or by efficient concussion.^[24] For many years, TNT used to be the reference point for the Figure of insensitivity. TNT had a rating of exactly 100 on the "F of I" scale. The reference has since been changed to a more sensitive explosive called RDX, which has an F of I rating of 80.^[25]

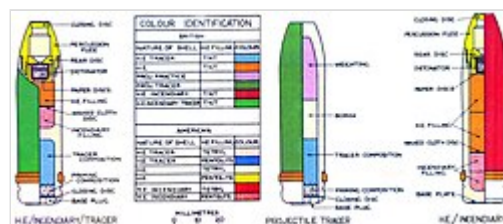
Energy content

The energy density of TNT is used as a reference point for many other explosives, including nuclear weapons, as their energy content is measured in equivalent tonnes (metric tons, t) of TNT. The energy used by NIST to define the equivalent is 4.184 GJ/t or exactly 1 kcal/g.^[26]

For safety assessments, it has been stated that the detonation of TNT, depending on circumstances, can release 2.673–6.702 GJ/t.^[27]

The heat of combustion however is 14.5 GJ/t (14.5 MJ/kg or 4.027 kWh/kg), which requires that the carbon in TNT fully react with atmospheric oxygen, which does not occur in the initial event.^[28]

For comparison, gunpowder contains 3 MJ/kg, dynamite contains 7.5 MJ/kg, and gasoline contains 47.2 MJ/kg (though gasoline requires an oxidant, so an optimized gasoline and O₂ mixture contains 10.4 MJ/kg).



Cross-sectional view of Oerlikon 20 mm cannon shells (dating from c. 1945) showing color codes for TNT and pentolite fillings

Detection

Various methods can be used to detect TNT, including optical and electrochemical sensors and explosive-sniffing dogs. In 2013, researchers from the Indian Institutes of Technology using noble-metal quantum clusters could detect TNT at the sub-zeptomolar (10^{-18} mol/m³) level.^[29]

Safety and toxicity

TNT is poisonous, and skin contact can cause skin irritation, causing the skin to turn a bright yellow-orange color. During the First World War, female munition workers who handled the chemical found that their skin turned bright yellow, which resulted in their acquiring the nickname "canary girls" or simply "canaries".^[30]

People exposed to TNT over a prolonged period tend to experience anemia and abnormal liver functions. Blood and liver effects, spleen enlargement and other harmful effects on the immune system have also been found in animals that ingested or breathed trinitrotoluene. There is evidence that TNT adversely affects male fertility.^[31] TNT is listed as a possible human carcinogen, with carcinogenic effects demonstrated in animal experiments with rats, although effects upon humans so far amount to none (according to IRIS of March 15, 2000).^[32] Consumption of TNT produces red urine through the presence of breakdown products and not blood as sometimes believed.^[33]

Some military testing grounds are contaminated with wastewater from munitions programs, including contamination of surface and subsurface waters which may be colored pink because of the presence of TNT. Such contamination, called "pink water", may be difficult and expensive to remedy.

TNT is prone to exudation of dinitrotoluenes and other isomers of trinitrotoluene when projectiles containing TNT are stored at higher temperatures in warmer climates. Exudation of impurities leads to formation of pores and cracks (which in turn cause increased shock sensitivity). Migration of the exudated liquid into the fuze screw thread can form *fire channels*, increasing the risk of accidental detonation. Fuze malfunction can also result from the liquid migrating into the fuze mechanism.^[34] Calcium silicate is mixed with TNT to mitigate the tendency towards exudation.^[35]

Pink and red water

Pink water is wastewater saturated with trinitrotoluene (at a concentration of up to 150 ppm).^[36] Pink water may be produced from equipment washing processes following munitions filling or demilitarization operations.^{[37][38]} Due to variances in the methods used at different production sites, pink water's composition is indefinite, and depends on the exact process used at a particular site; the runoff from a Munitions factory may also contain RDX or HMX, depending on the mix used.^[37]

Red water is wastewater formed when crude trinitrotoluene is reacted with an aqueous sodium sulfite solution (referred to as "sellite"). Sellite reacts with the undesired components to form water soluble compounds, but leaves the desired isomer unchanged. The water used to wash these now-soluble waste

compounds away is referred to as red water, or sometimes "sellite water" It has a complex composition containing more than a dozen aromatic compounds, but the principal components are inorganic salts (sodium sulfate, sodium sulfite, sodium nitrite and sodium nitrate) and sulfonated nitroaromatics.^[36]

Pink and red water are colorless at the time of generation; the color is produced by photolytic reactions under the influence of sunlight. Despite the names, red and pink water are not necessarily different shades; the color depends mainly on the duration of solar exposure. If exposed long enough, "pink" water may turn various shades of pink, red, rusty orange, or black.^{[38][39]}

Because of the toxicity of TNT, the discharge of pink water to the environment has been prohibited in the US and many other countries for decades, but ground contamination may exist in very old plants. However, RDX and tetryl contamination is usually considered more problematic, as TNT has very low soil mobility. Red water is significantly more toxic and as such it has always been considered hazardous waste. It has traditionally been disposed of by evaporation to dryness (as the toxic components are not volatile), followed by incineration. Much research has been conducted to develop better disposal processes.

Ecological impact

Because of its suitability in munitions, TNTs toxicity has been characterized and reported. Residual TNT from manufacture, storage, and use can pollute water, soil, the atmosphere, and the biosphere.^[40]

The concentration of TNT in contaminated soil can reach 50 g/kg of soil, where the highest concentrations can be found on or near the surface. In September 2001, the United States Environmental Protection Agency (USEPA) declared TNT a pollutant whose removal is a priority.^[41] The USEPA maintains that TNT levels in soil should not exceed 17.2 milligrams per kilogram of soil and 0.01 milligrams per litre of water.^[42]

Aqueous solubility

Dissolution is a measure of the rate that solid TNT in contact with water is dissolved. The relatively low aqueous solubility of TNT causes solid particles to be continuously released to the environment over extended periods of time.^[43] Studies have shown that TNT dissolves more slowly in saline water than in freshwater. However, when salinity is altered, TNT dissolves at the same speed.^[44] Because TNT is moderately soluble in water, it can migrate through subsurface soil, and cause groundwater contamination.^[45]

Soil adsorption

TNT and its transformation products are known to adsorb to surface soils and sediments, where they undergo reactive transformation or remained stored.^[46] The movement of organic contaminants through soils is a function of their ability to associate with the mobile phase (water) and a stationary phase (soil). Materials that associate strongly with soils move slowly through soil. The association constant for TNT with soil is 2.7 to 11 L/kg of soil.^[47] This means that TNT has a one- to tenfold tendency to adhere to soil particulates than not when introduced into the soil.^[43] Hydrogen bonding and ion exchange are two suggested mechanisms of adsorption between the nitro functional groups and soil colloids.

The adsorption of TNT is sensitive to soil type.^[48]

Additional studies have shown that the mobility of TNT degradation products is likely to be lower "than TNT in subsurface environments where specific adsorption to clay minerals dominates the sorption process."^[48] Thus, the mobility of TNT and its transformation products are dependent on the characteristics of the sorbent.^[48] The mobility of TNT in groundwater and soil has been extrapolated from "sorption and desorption isotherm models determined with humic acids, in aquifer sediments, and soils".^[48] From these models, it is predicted that TNT has a low retention and transports readily in the environment.^[41]

Compared to other explosives, TNT has a higher association constant with soil, meaning it adheres more with soil than with water. Conversely, other explosives, such as RDX and HMX with low association constants (ranging from 0.06 to 7.3 L/kg and 0 to 1.6 L/kg respectively) can move more rapidly in water.^[43]

Chemical breakdown

The transformation of TNT is significantly enhanced under anaerobic conditions as well as under highly reducing conditions. TNT transformations in soils can occur both biologically and abiotically.^[48]

Biodegradation

While some bacteria and fungi can incorporate and transform TNT into various aromatic compounds, biomineralization of large amounts of TNT by soil organisms has yet to be reported.^[49] Wild and transgenic plants have also demonstrated the ability to phytoremediate TNT by absorption from soil and water. The primary mechanism of degradation within plants is believed to be via the enzyme nitroreductase .^[50]

See also

- Dynamite
- Environmental fate of TNT
- IMX-101
- List of explosives used during World War II
- Phlegmatized
- RE factor
- Table of explosive detonation velocities
- TNT equivalent
- Webster's test

References

1. "Trinitrotoluene" (<https://pubchem.ncbi.nlm.nih.gov/compound/8376#section=IUPAC-Name>).
2. 2,4,6-Trinitrotoluene (<http://www.inchem.org/documents/icsc/icsc/eics0967.htm>). inchem.org
3. NIOSH Pocket Guide to Chemical Hazards. "#0641" (<https://www.cdc.gov/niosh/npg/npgd0641.html>). National Institute for Occupational Safety and Health (NIOSH).

4. "2,4,6-Trinitrotoluene" (<https://www.cdc.gov/niosh/idlh/118967.html>). *Immediately Dangerous to Life or Health Concentrations*. National Institute for Occupational Safety and Health.
5. "trinitrotoluene" (<https://www.dictionary.com/browse/trinitrotoluene>). *Dictionary.com Unabridged* (Online). n.d.
"trinitrotoluene" (<https://www.merriam-webster.com/dictionary/trinitrotoluene>). *Merriam-Webster.com Dictionary*. Merriam-Webster. OCLC 1032680871 (<https://search.worldcat.org/oclc/1032680871>).
6. Wilbrand, J. (1861). "Notiz über Trinitrotoluol" (<https://books.google.com/books?id=qmgTAAAAQAAJ&pg=PA178>). *Annalen der Chemie und Pharmacie*. **128** (2): 178–179. doi:10.1002/jlac.18631280206 (<https://doi.org/10.1002%2Fjac.18631280206>).
7. Peter O. K. Krehl (2008). *History of Shock Waves, Explosions and Impact: A Chronological and Biographical Reference* (<https://books.google.com/books?id=PmuqCHDC3pwC&pg=PA404>). Springer Science & Business Media. p. 404. ISBN 978-3-540-30421-0.
8. Brown GI (1998). *The Big Bang: a History of Explosives* (<https://archive.org/details/bigbanghistoryof00brow/page/151>). Sutton Publishing. pp. 151–153 (<https://archive.org/details/bigbanghistoryof00brow/page/151>). ISBN 978-0-7509-1878-7.
9. Norman Skentelbery (1975). *Arrows to Atom Bombs: A History of the Ordnance Board* (<https://books.google.com/books?id=ySQGAAAAMAAJ>) (2nd ed.). Ordnance Board. p. 99.
10. Fairfield AP (1921). *Naval Ordnance*. Lord Baltimore Press. pp. 49–52.
11. Urbanski T (1964). *Chemistry and Technology of Explosives*. Vol. 1. Pergamon Press. pp. 389–91. ISBN 978-0-08-010238-2.
12. Nash C, Nelson T, Stewart J, Carper W (1989). "Molecular structure and vibrational analysis of 2,4,6-trinitrotoluene and 2,4,6-trinitrotoluene- α - d_3 ". *Spectrochimica Acta Part A*. **45** (5): 585–588. Bibcode:1989AcSpA..45..585N (<https://ui.adsabs.harvard.edu/abs/1989AcSpA..45..585N>). doi:10.1016/0584-8539(89)80010-8 (<https://doi.org/10.1016%2F0584-8539%2889%2980010-8>).
13. Miller JS, Johansen RT (1976). "Fracturing Oil Shale with Explosives for In Situ Recovery" (https://web.archive.org/web/20181002204137/https://web.anl.gov/PCS/acsfuel/preprint%20archive/Files/19_2_LOS%20ANGELES_04-74_0060.pdf) (PDF). *Shale Oil, Tar Sand and Related Fuel Sources*: 151. Bibcode:1976sots.rept...98M (<https://ui.adsabs.harvard.edu/abs/1976sots.rept...98M>). Archived from the original (https://web.anl.gov/PCS/acsfuel/preprint%20archive/Files/19_2_LOS%20ANGELES_04-74_0060.pdf) (PDF) on 2 October 2018. Retrieved 27 March 2015.
14. "TNT" (https://www.ch.ic.ac.uk/vchemlib/mim/bristol/tnt/tnt_text.htm). *www.ch.ic.ac.uk*. Retrieved 2022-02-28.
15. Campbell J (1985). *Naval weapons of World War Two*. London: Conway Maritime Press. p. 100. ISBN 978-0-85177-329-2.
16. *U.S. Explosive Ordnance, Bureau of Ordnance* (<http://www.maritime.org/doc/ordnance/index.htm>). Washington, D.C.: U.S. Department of the Navy. 1947. p. 580.
17. United States Senate Committee on the Judiciary (1976). *Control of Explosives – Administration and Execution of the Laws Pertaining to the Control of Explosives* (<https://books.google.com/books?id=JN8HauvrqT8C&pg=PA157>). Washington: US Government Printing Office. pp. 157–159. Retrieved 11 September 2025.
18. Military Specification MIL-C-401
19. Cooper PW (1996). *Explosives Engineering*. Wiley-VCH. ISBN 978-0-471-18636-6.
20. [secondary source] *webpages:submarine torpedo explosive* (http://www.dutchsubmarines.com/specials/special_torpedoes_mines.htm) *Deprecated link archived 2013-01-02 at archive.today* Retrieved 2011-12-02

21. *scribd.com* website showing copy of a North American Intelligence document see: page 167 (<https://www.scribd.com/doc/39682972/Encyclopedia-of-Explosives-A-Compilation-of-Principal-Explosives-Their-Characteristics-Processes-of-Manufacture-And-Uses-USA-1960>) Archived (<https://web.archive.org/web/20130510223600/http://www.scribd.com/doc/39682972/Encyclopedia-of-Explosives-A-Compilation-of-Principal-Explosives-Their-Characteristics-Processes-of-Manufacture-And-Uses-USA-1960>) 2013-05-10 at the *Wayback Machine* Retrieved 2011-12-02
22. Furman D, Kosloff R, Dubnikova F, Zybin SV, Goddard WA, Rom N, Hirshberg B, Zeiri Y (6 March 2014). "Decomposition of Condensed Phase Energetic Materials: Interplay between Uni- and Bimolecular Mechanisms" (<https://resolver.caltech.edu/CaltechAUTHORS:20140428-125540687>). *Journal of the American Chemical Society*. **136** (11). American Chemical Society (ACS): 4192–4200. Bibcode:2014JChS.136.4192F (<https://ui.adsabs.harvard.edu/abs/2014JChS.136.4192F>). doi:10.1021/ja410020f (<https://doi.org/10.1021%2Fja410020f>). ISSN 0002-7863 (<https://search.worldcat.org/issn/0002-7863>). PMID 24495109 (<https://pubmed.ncbi.nlm.nih.gov/24495109>).
23. Brown GI (1998). *The Big Bang: A History of Explosives*. Sutton Publishing. pp. 158–163. ISBN 0-7509-1878-0.
24. O'Neil, Smith A, Heckelman PE, Budavari S (15 October 2001). *Merck Index* (13th ed.). Wiley. ISBN 9780911910131. 9801.
25. Raha K, Chhabra J (1993). "Static charge development and impact sensitivity of high explosives" (<https://linkinghub.elsevier.com/retrieve/pii/030438949385100S>). *Journal of Hazardous Materials*. **34** (3): 385–391. Bibcode:1993JHzM...34..385R (<https://ui.adsabs.harvard.edu/abs/1993JHzM...34..385R>). doi:10.1016/0304-3894(93)85100-S (<https://doi.org/10.1016%2F0304-3894%2893%2985100-S>).
26. "Guide for the Use of the International System of Units (SI)" (<https://www.nist.gov/physical-measurement-laboratory/nist-guide-si-appendix-b8>). *NIST*. 2008. Retrieved 2024-03-25.
27. "Blast effects of external explosions (Section 4.8 Limitations of the TNT equivalent method)" (<https://web.archive.org/web/20160810225249/http://hal.archives-ouvertes.fr/hal-00629253/document>). 2011. p. 16. Archived from the original (<https://hal.archives-ouvertes.fr/hal-00629253/document>) on 2016-08-10.
28. Babrauskas V (2003). *Ignition Handbook*. Issaquah, WA: Fire Science Publishers/Society of Fire Protection Engineers. p. 453. ISBN 978-0-9728111-3-2.
29. Grad P (April 2013). "Quantum clusters serve as ultra-sensitive detectors" (<http://www.chemengonline.com/quantum-clusters-serve-as-ultra-sensitive-detectors/>). *Chemical Engineering*.
30. "The Canary Girls: The workers the war turned yellow" (<https://www.bbc.com/news/uk-england-39434504>). *BBC News*. 2017-05-20. Retrieved 2021-02-07.
31. Toxicological Profile for 2,4,6-Trinitrotoluene (<https://www.atsdr.cdc.gov/toxprofiles/tp81.pdf>). *atsdr.cdc.gov*
32. "2,4,6-Trinitrotoluene" (<https://pubchem.ncbi.nlm.nih.gov/compound/8376>). *www.nlm.nih.gov*.
33. "2,4,6-Trinitrotoluene" (<https://www.atsdr.cdc.gov/ToxProfiles/tp81-c2.pdf>) (PDF). Agency for Toxic Substances and Disease Registry. Retrieved 2010-05-17.
34. Akhavan J (2004). *The Chemistry of Explosives* (<https://books.google.com/books?id=9tIQDn2uZz4C&pg=PA11>). Royal Society of Chemistry. pp. 11–. ISBN 978-0-85404-640-9.
35. "Explosive & Propellant Additives" (https://web.archive.org/web/20160304032754/http://www.islandgroup.com/military/explosives_additives.php). *islandgroup.com*. Archived from the original (http://www.islandgroup.com/military/explosives_additives.php) on 2016-03-04. Retrieved 2014-06-07.
36. Yinon J (1990). *Toxicity and metabolism of explosives* (<https://books.google.com/books?id=BD3c7FN4x5YC&q=%22red+water%22+tnt&pg=PA39>). CRC Press. pp. 38–39. ISBN 0-8493-5128-6.

37. National Defense Center for Environmental Excellence. Pink Water Treatment Options (<https://apps.dtic.mil/sti/pdfs/ADA295802.pdf>) (PDF) (Report). U.S. Army Environmental Center. Retrieved 2024-04-25.
38. Van Deuren J, Lloyd T, Chhetry S, Liou R, Peck J (January 2002). Remediation Technologies Screening Matrix and Reference Guide (https://www.frtr.gov/matrix2/section2/2_10_3.html) (Report) (4 ed.). U.S. Army Environmental Center. Retrieved 2024-04-25.
39. Burlinson NE, Kaplan LA, Adams CE (1973-10-03). Photochemistry of TNT: Investigation of the "Pink Water" Problem (<https://apps.dtic.mil/sti/pdfs/AD0769670.pdf>) (PDF) (Report). U.S. Army Environmental Center. Retrieved 2024-04-25.
40. Ascend Waste and Environment (7 June 2015). The health and environmental impacts of hazardous wastes: Impact Profiles (<https://www.environment.gov.au/system/files/resources/bc0e52ba-8f78-4ce1-83b4-4910f4a1f0e9/files/hazardous-waste-impacts.pdf>) (PDF). awe.gov.au (Report). Retrieved 22 April 2022.
41. Esteve-Núñez A, Caballero A, Ramos JL (2001). "Biological degradation of 2,4,6-trinitrotoluene" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC99030>). *Microbiol. Mol. Biol. Rev.* **65** (3): 335–52, table of contents. doi:10.1128/MMBR.65.3.335-352.2001 (<https://doi.org/10.1128/MMBR.65.3.335-352.2001>). PMC 99030 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC99030>). PMID 11527999 (<https://pubmed.ncbi.nlm.nih.gov/11527999>).
42. Ayoub K, van Hullebusch ED, Cassir M, Bermond A (2010). "Application of advanced oxidation processes for TNT removal: A review". *J. Hazard. Mater.* **178** (1–3): 10–28. doi:10.1016/j.jhazmat.2010.02.042 (<https://doi.org/10.1016/j.jhazmat.2010.02.042>). PMID 20347218 (<https://pubmed.ncbi.nlm.nih.gov/20347218>).
43. Pichte J (2012). "Distribution and Fate of Military Explosives and Propellants in Soil: A Review" (<https://doi.org/10.1155/2012/2F617236>). *Applied and Environmental Soil Science*. **2012** (1): 1–33. Bibcode:2012ApESS201217236P (<https://ui.adsabs.harvard.edu/abs/2012ApESS201217236P>). doi:10.1155/2012/617236 (<https://doi.org/10.1155/2012/617236>).
44. Brannon JM, Price CB, Yost SL, Hayes C, Porter B (2005). "Comparison of environmental fate and transport process descriptors of explosives in saline and freshwater systems". *Mar. Pollut. Bull.* **50** (3): 247–51. Bibcode:2005MarPB..50..247B (<https://ui.adsabs.harvard.edu/abs/2005MarPB..50..247B>). doi:10.1016/j.marpolbul.2004.10.008 (<https://doi.org/10.1016/j.marpolbul.2004.10.008>). PMID 15757688 (<https://pubmed.ncbi.nlm.nih.gov/15757688>).
45. Halasz A, Groom C, Zhou E, Paquet L, Beaulieu C, Deschamps S, Corriveau A, Thiboutot S, Ampleman G, Dubois C, Hawari J (2002). "Detection of explosives and their degradation products in soil environments". *J Chromatogr A*. **963** (1–2): 411–8. doi:10.1016/S0021-9673(02)00553-8 ([https://doi.org/10.1016/S0021-9673\(02\)00553-8](https://doi.org/10.1016/S0021-9673(02)00553-8)). PMID 12187997 (<https://pubmed.ncbi.nlm.nih.gov/12187997>).
46. Douglas TA, Johnson L, Walsh M, Collins C (2009). "A time series investigation of the stability of nitramine and nitroaromatic explosives in surface water samples at ambient temperature" (<https://zenodo.org/record/851720>). *Chemosphere*. **76** (1): 1–8. Bibcode:2009Chmsp..76....1D (<https://ui.adsabs.harvard.edu/abs/2009Chmsp..76....1D>). doi:10.1016/j.chemosphere.2009.02.050 (<https://doi.org/10.1016/j.chemosphere.2009.02.050>). PMID 19329139 (<https://pubmed.ncbi.nlm.nih.gov/19329139>).
47. Haderlein SB, Weissmahr KW, Schwarzenbach RP (January 1996). "Specific Adsorption of Nitroaromatic Explosives and Pesticides to Clay Minerals". *Environmental Science & Technology*. **30** (2): 612–622. Bibcode:1996EnST...30..612H (<https://ui.adsabs.harvard.edu/abs/1996EnST...30..612H>). doi:10.1021/es9503701 (<https://doi.org/10.1021/es9503701>).
48. Pennington JC, Brannon JM (February 2002). "Environmental fate of explosives". *Thermochimica Acta*. **384** (1–2): 163–172. Bibcode:2002TcAc..384..163P (<https://ui.adsabs.harvard.edu/abs/2002TcAc..384..163P>). doi:10.1016/S0040-6031(01)00801-2 ([https://doi.org/10.1016/S0040-6031\(01\)00801-2](https://doi.org/10.1016/S0040-6031(01)00801-2)).

49. Hawari J, Beaudet S, Halasz A, Thiboutot S, Ampleman G (2000). "Microbial degradation of explosives: biotransformation versus mineralization". *Appl. Microbiol. Biotechnol.* **54** (5): 605–18. doi:10.1007/s002530000445 (<https://doi.org/10.1007/s002530000445>). PMID 11131384 (<https://pubmed.ncbi.nlm.nih.gov/11131384>). S2CID 22362850 (<https://api.semanticscholar.org/CorpusID:22362850>).
50. G. Adamia, M. Ghoghoberidze, D. Graves, G. Khatishashvili, G. Kvesitadze, E. Lomidze, D. Ugrekhelidze, G. Zaalishvili (2006). "Absorption, distribution, and transformation of TNT in higher plants" (<https://www.sciencedirect.com/science/article/abs/pii/S0147651305001193>). *Ecotoxicology and Environmental Safety*. **64** (2): 136–145 – via Elsevier Science Direct.

External links

- [Dynamite and TNT \(http://www.periodicvideos.com/videos/mv_dynamite.htm\)](http://www.periodicvideos.com/videos/mv_dynamite.htm) at *The Periodic Table of Videos* (University of Nottingham)
 - "TNT" on YouTube (<https://www.youtube.com/watch?v=vb3y6m-buco>) showing the shockwave and typical black smoke cloud from detonation of 160 kilograms of pure TNT
 - CDC – NIOSH Pocket Guide to Chemical Hazards (<https://www.cdc.gov/niosh/npg/npgd0641.html>)
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